BFKL evolution of the projectile: all terms $\propto \log \frac{\vee}{k^+}$ from your `BFKL.tex` and `Other_terms.tex`, with $\vee$ exact and no large-p^+ step

Notation as in your files; $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $U^{bc} = U^{cb}$. In the rows: black = your `BFKL.tex`, blue = added (the $\log \frac{\vee}{k^+}$ still inside `Other_terms.tex`), red = to be removed. The remaining terms are in `Remaining_rows.pdf` and `Rest_terms.pdf`.

1 Answer

Rule. The only constraints are $\vee > k^+$ and $p^+ > \Lambda$. $\Lambda \to 0$ wherever possible; $\vee$ is exact. Only explicit logarithms are split, and exactly:

$$\log\left(\frac{\vee}{k^+} - 1\right) = \log\frac{\vee}{k^+} + \log\left(1 - \frac{k^+}{\vee}\right), \qquad \log\frac{(\vee - k^+)A + k^+B}{\Lambda A + k^+ B} = \log\frac{\vee}{k^+} + \log\frac{(1-\frac{k^+}{\vee})A + \frac{k^+}{\vee}B}{B} + O(\Lambda), \qquad \dots$$

($A = (x-z)^2$, $B = (x-w)^2$; the full table is Sec. 2 of `LL_terms.pdf`). Every p^+ and ξ integral is kept as you wrote it: $\left[\frac{1}{p^+}\right]_+$, $\frac{\xi}{(1-\xi)_+}$, upper limit $\vee - k^+$, lower limit $\frac{k^+}{\vee}$. There is no large-p^+ (small-ξ) step, so these integrals give no BFKL term.

Withdrawn from the previous version.

- The terms (B) of Group III Rows III (both parts), V, VI, Eq. (7.1), and H_4 of Group II Row IV. Both came from the region of large p^+ (resp. $\xi \to \frac{k^+}{\vee}$).
- The subtractions $-$(B) and $-\log\frac{\vee}{k^+}H_4$ in the remaining terms.
- The rule that $\log(\frac{\vee}{k^+} - 1)$ in line 2 of these rows is not a BFKL term. It relied on the same region.
- `Why_log_vee.pdf`.

Result.

- **Your two files are complete.** Row by row, `BFKL.tex` (without the red lines) + `Other_terms.tex` = the row minus its $\log\frac{k^+}{\Lambda}$ (JIMWLK) part, numerically. The differences, $\le 2.5\times10^{-9}$ at $\Lambda = 10^{-9}$, are the $O(\Lambda)$ remainders. The terms with an unevaluated p^+, ξ or momentum integral are identical to those of your Group files.
- **Remove three lines** (red):
  - Group I Row V 2b UV: Row V has no $\vee$; this line comes from the missing factor 2 after "Setting $\Lambda \to 0$" (E1).
  - Group II Row IV, both lines: $-\log\frac{\vee}{k^+}\times$(line 2) and $\log\frac{\vee}{k^+}H_4$. They are statements about the ξ integral of line 1, not logarithms of the row.

  With these lines, `BFKL.tex` + `Other_terms.tex` exceeds the row (by ≥ 0.66 relative).
- **Add** (blue) every $\log\frac{\vee}{k^+}$ that is still in `Other_terms.tex`:
  - Group I Rows III and IV;
  - Group II Row I (both parts) and Row IV (line 2);
  - Group III Rows I, II, III (both parts), V, VI (line 2).

  Then the list is exactly the $\log\frac{\vee}{k^+}$ part of your rows (to 4.4×10^{-17}).
- **What this list gives.**
  - **Three ρ's** (35 terms, Sec. 6): they **do not cancel**: the residual is up to 1.01 times the largest term (three configurations). It is exactly the sum of the three blue line-2 terms of Group III Rows III(1), V, VI; all other terms cancel among themselves, to 3.6×10^{-16}. Your `BFKL.tex` alone leaves 0.39, and the three line-2 terms of Group III Rows III, V, VI alone leave 1.11.

– **Two ρ's: not BFKL⊗LO.** The previous list had (B) and H_4 instead of the line-2 terms, and its two-ρ terms added up to BFKL⊗LO. The change is +(the line-2 terms of Group III Rows III, V, VI and Group II Row IV) −(B) $-H_4$. Its two-ρ terms do not cancel: they are non-zero in all 41 kernel structures in which they appear (32 from the line-2 terms, 9 from (B) and H_4; Sec. 7).

– **Pole.** The blue term of Group II Row IV is the only term of the list with $\frac{1}{\epsilon}$. Its pole equals that of the red Group I Row V line (to 1.8×10^{-15}), so nothing in the list cancels it (Sec. 8).

– Your commented "BFKL contribution" (your App. E formula, BFKL⊗LO) is the sum of the previous list, not of this one.

So, with $\vee$ exact and the p^+, ξ integrals kept as written, the coefficient of the explicit $\log\frac{\vee}{k^+}$ is *not* BFKL⊗LO.

The rows themselves are unchanged (check A). Only the bookkeeping changes. The rest of their $\vee$ dependence is in the remaining terms:

- the p^+ integrals up to $\vee - k^+$ (line 1 of Group III Rows III, V, VI);
- the ξ integrals from $\frac{k^+}{\vee}$ (lines 1 and 3 of Group II Row IV);
- $\log(1-\frac{k^+}{\vee})$ and $\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$;
- R^i of Group I Row II.

A second exact bookkeeping (for your decision). The split above is not the only exact one. In the subtraction of $\left[\frac{1}{p^+}\right]_+$ one may use $\frac{1}{p^+}=\frac{1}{p^++k^+}+\frac{k^+}{p^+(p^++k^+)}$. With $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$ this gives, exactly,

$$\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+f(p^+)=\int_0^{\vee-k^+}dp^+\,\frac{k^+\,[f(p^+)-f(0)]}{p^+(p^++k^+)}+\int_0^{\vee-k^+}dp^+\,\frac{f(p^+)}{p^++k^+}-f(0)\log\frac{\vee}{k^+}.$$

For $\frac{\xi}{(1-\xi)_+}$ the analogue is $\frac{1}{\xi(1-\xi)}=\frac{1}{\xi}+\frac{1}{1-\xi}$, which gives $-g(1)\log\frac{\vee}{k^+}$.

- Your line 2 is $f(0)\big[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\big]$ (resp. with $g(1)$), so the explicit $-f(0)\log\frac{\vee}{k^+}$ cancels its $\log\frac{\vee}{k^+}$ exactly. Group III Rows III, V, VI and Group II Row IV then have no BFKL term at all.
- Your red line 1 of Group II Row IV is exactly this $-g(1)\log\frac{\vee}{k^+}$. It belongs to this bookkeeping, with line 1 of `Other_terms.tex` rewritten accordingly.

Both bookkeepings are exact; they differ only in which $\vee$ dependence is called an explicit logarithm:

	this file (explicit logarithms only)	second bookkeeping (subtraction split)
BFKL term of G3 III, V, VI, G2 IV	$\log\frac{\vee}{k^+}$ part of line 2	none
three-ρ terms	do not cancel (1.01)	cancel (3.6×10^{-16})
two-ρ terms vs BFKL⊗LO	differ in 41 structures	differ in 9 structures (−(B), $-H_4$)
pole $\times\log\frac{\vee}{k^+}$ in the list	yes (G2 IV)	none

In neither is the two-ρ sum BFKL⊗LO. That needs (B) and H_4, i.e. the large-p^+ region of the integrals that both keep unevaluated.

2 Group I (with c.c.)

2.1 Row I

Only four ρ's (Sec. 5).

2.2 Row II

Row II, first part. No BFKL term. The only $\log\frac{\vee}{k^+}$ is in $\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$, whose coefficient depends on $\vee$. For every $\vee>k^+$ it is bounded, $0\le-\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}\le\log^2 2$, with the maximum at $\vee=2k^+$. It stays in the remaining terms, as in your `Other_terms.tex`.

Row II, second part, d^3N_{2a}. As you have it. Your `BFKL.tex`:

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\Bigg[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\\&-\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\Bigg]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\\&-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'}e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\Bigg[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\Bigg]\times\\&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\end{aligned}\tag{2.1}$$

Row II, second part, d^3N_{2b}. No BFKL term (R^i depends on $\vee$ only through $\frac{k^+}{\vee}$).

2.3 Row III

Row III, d^3N_1. Your line is correct. Added: lines 1, 2 and 4 of d^3N_1 in `Other_terms.tex` still contain $\log\frac{\vee}{k^+}$: $\log(\frac{\vee}{k^+}-1) = \log\frac{\vee}{k^+} + \log(1-\frac{k^+}{\vee})$, $\log\frac{\vee(x-z)^2-k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2-k^+(x-w)^2} = \log\frac{\vee}{k^+} + \ldots$, $\log\frac{k^+(x-w)^2-\vee(x-z)^2}{k^+(x-w)^2-\Lambda(x-z)^2} = \log\frac{\vee}{k^+} + \ldots$. Your `BFKL.tex`:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \tag{2.2}$$

From `Other_terms.tex`:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ &\\
&-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left(\frac{\vee}{k^+}\right)\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left(\frac{\vee}{k^+}\right)\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}\end{aligned} \tag{2.3}$$

Row III, d^3N_2. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of all eight lines of d^3N_2 in `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \tag{2.4}$$

From `Other_terms.tex`:

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ &\\
&\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\end{aligned}$$

$$
-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}
$$

$$
\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\} \tag{2.5}
$$

Row III, d^3N_{3a}. No BFKL term.

Row III, d^3N_{3b}. No BFKL term.

2.4 Row IV

Row IV. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of $\log\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}$ and $\log(\frac{\vee}{k^+}-1)$ (both lines in `Other_terms.tex`). Your `BFKL.tex`:

$$
\frac{d^3N_{\rm NLO}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times
$$

$$
\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\right] \tag{2.6}
$$

From `Other_terms.tex`:

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\Bigg\{
$$

$$
\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]
$$

$$
+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times
$$

$$
\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\right] \tag{2.7}
$$

2.5 Row V

Row V, first part. No BFKL term.

Row V, second part, d^3N_{2a}. No BFKL term.

Row V, second part, $d^3N_{2b\,\rm UV}$. Remove (red). Row V has $0<p^+<k^+$: no $\vee$ and no $\log\frac{\vee}{k^+}$. Before "Setting $\Lambda\to0$" your bracket is $2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}=2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}$. The $2\log\frac{k^+}{\Lambda}$ is in your JIMWLK terms, and $2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}$ is in `Other_terms.tex`. This line comes from the bracket $\left[\log\frac{\vee}{k^+}+\log\frac{k^+}{\Lambda}\right]-2\log\frac{\vee}{k^+}$ after "Setting $\Lambda\to0$", which lacks a factor 2 on the first bracket (E1). Your `BFKL.tex`:

$$
\frac{d^3N_{2b\,\rm UV\,NLO}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu_{\overline{MS}}^2}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}
$$

$$
\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{2.8}
$$

Row V, second part, $d^3N_{2b\,\rm finite}$. No BFKL term.

3 Group II

3.1 Row I

Row I, first part, first line. Not in your `BFKL.tex`. Added: the $\log\frac{\vee}{k^+}$ parts of all three lines (`Other_terms.tex`). From `Other_terms.tex`:

$$
\frac{d^3N_{1\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]
$$

$$
+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}
$$

$$
- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}
$$

$$
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.1}
$$

Row I, first part, second line. Added: the log $\frac{\vee}{k^+}$ parts of all three lines. From `Other_terms.tex`:

$$
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]
$$

$$
-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}
$$

$$
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}
$$

$$
\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{3.2}
$$

Row I, second part, d^3N_{2a}. No BFKL term: $\vee$ cancels between numerator and denominator of the first logarithm.

Row I, second part, d^3N_{2b}. Not in your `BFKL.tex`; all three lines are added. From `Other_terms.tex`:

$$
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left(\frac{\vee}{k^+}\right)\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left(\frac{\vee}{k^+}\right)
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \tag{3.3}
$$

Row I, second part, d^3N_{2c}**.** Not in your `BFKL.tex`; both lines are added. From `Other_terms.tex`:

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
& \left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
& \left. + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right] \times \log\left(\frac{\vee}{k^+}\right) \\
& \Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
& - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \\
& - \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
& \left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
& \left. + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right] \times \log\left(\frac{\vee}{k^+}\right) \\
& \Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
& - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.4}
$$

Row I, second part, d^3N_{2d}**.** Your line is correct. Added: the two lines of `Other_terms.tex`. Your `BFKL.tex`:

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = {}& -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]} \\
& \left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
& \left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right] \\
& \Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
& - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.5}
$$

From `Other_terms.tex`:

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = {}& \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
& \left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
& \left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
& \log\left(\frac{\vee}{k^+}\right)\times \\
& \Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x) \\
& - f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \\
& + \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
& \left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right. \\
& \left. + \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times \\
& \log\left(\frac{\vee}{k^+}\right)\times
\end{aligned}
$$

$$\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.6}$$

Row I, second part, d^3N_{2e}. Your line is correct. Added: the two lines of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big] \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2-(x'-z)^2]}$$

$$\Big[\frac{(x-z)\cdot(z-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{4\,(x-w)^2\,(z-w')^2}$$

$$+ \frac{(x'-z)\cdot(z-w')\,\big[(x-z)^2-(x-w)^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\big[(x-z)^2-(x-w)^2\big]}{4\,(z-w)^2\,(x'-w')^2} \Big]$$

$$\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.7}$$

From `Other_terms.tex`:

$$\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$

$$\Big[\frac{(x-z)\cdot(z-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{4\,(x-w)^2\,(z-w')^2}$$

$$+ \frac{(x'-z)\cdot(z-w')\,\big[(x-z)^2-(x-w)^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\big[(x-z)^2-(x-w)^2\big]}{4\,(z-w)^2\,(x'-w')^2} \Big] \times$$

$$\log\Big(\frac{\vee}{k^+}\Big) \times$$

$$\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]$$

$$- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$

$$\Big[\frac{(x-z)\cdot(z-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\big[(x'-z)^2-(x'-w')^2\big]}{4\,(x-w)^2\,(z-w')^2}$$

$$+ \frac{(x'-z)\cdot(z-w')\,\big[(x-z)^2-(x-w)^2\big]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\big[(x-z)^2-(x-w)^2\big]}{4\,(z-w)^2\,(x'-w')^2} \Big] \times$$

$$\log\Big(\frac{\vee}{k^+}\Big) \times$$

$$\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.8}$$

Row I, second part, d^3N_{2f}. Your line is correct. Added: lines 1 and 3 of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\Big[\frac{\vee}{k^+}\Big]\,\frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{\big[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\big]}{(z-w)^2\,(z-w')^2\,[(x-z)^2-(x-w)^2]\,[(x'-z)^2-(x'-w')^2]}$$

$$\Big[f^{c'bd'}\, U^{ac'}(w')\, \rho^{d'}(x') - f^{ac'd'}\, U^{bc'}(z) U^{e'd'}(x')\, \rho^{e'}(x') \Big] \times \Big[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x) \Big] \tag{3.9}$$

From `Other_terms.tex`:

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,[(x-z)^2-(x-w)^2]}\, \frac{[(x'-z)^2-(x'-w')^2]}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \times$$

$$\log\Big(\frac{\vee}{k^+}\Big) \times$$

$$\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x') \rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x') \rho^{e}(x)$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]$$

$$+\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}$$
$$\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left(\frac{\vee}{k^+}\right)\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{3.10}$$

3.2 Row II

Only four ρ's (Sec. 5).

3.3 Row III

Only four ρ's (Sec. 5).

3.4 Row IV

Row IV. Both lines: remove (red). Neither is a logarithm of your row: with them, `BFKL.tex` + `Other_terms.tex` exceeds the row by exactly these two lines (check A, Sec. 9). Line 1, $-\log\frac{\vee}{k^+}\times$(line 2 of `Other_terms.tex`), is the $\frac{d\xi}{\xi}$ part of the subtraction in $\frac{\xi}{(1-\xi)_+}$ of line 1 of `Other_terms.tex`, $-e^{-ik\cdot(y'-y)}\int_{k^+/\vee}^1\frac{d\xi}{\xi}$. It is not taken out there: the ξ integral is kept as you wrote it, so it would be counted twice. Line 2 (H_4) is what the region $y'\to y$ of that ξ integral gives for $\vee\gg k^+$; with $\vee$ exact it is not there. Added: the $\log\frac{\vee}{k^+}$ part of $\log(\frac{\vee}{k^+}-1)$ in line 2 of `Other_terms.tex`. It is the BFKL term of this row, and it carries the pole bracket (Sec. 8). Your `BFKL.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k}=-\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3.11}$$

From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k}=+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3.12}$$

4 Group III (with c.c.)

4.1 Row I

Row I. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of both lines of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$\Big[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\Big]\times$$
$$\Big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\Big] \tag{4.1}$$

From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$

$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.2}$$

4.2 Row II

Row II. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of both lines of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^{b}(y)\rho^{b'}(x')\right]\times$$
$$\left.\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]\right\} \tag{4.3}$$

From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\left.\right\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^{b}(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.4}$$

4.3 Row III

Row III, first part. Not in your `BFKL.tex`. Added: the $\log\frac{\vee}{k^+}$ part of $\log(\frac{\vee}{k^+}-1) = \log\frac{\vee}{k^+} + \log(1-\frac{k^+}{\vee})$ in line 2 of `Other_terms.tex`. Line 1, the p^+ integral up to $\vee - k^+$ with $\left[\frac{1}{p^+}\right]_+$, is kept as you wrote it. It contains no explicit logarithm and gives no BFKL term. From `Other_terms.tex`:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^{c}(y),\rho^{b}(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^{d}(y),\rho^{e}(x)\}\right] \tag{4.5}$$

Row III, second part. Added: the $\log\frac{\vee}{k^+}$ part of line 2, as in the first part. From `Other_terms.tex`:

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.6}$$

4.4 Row IV

Only four ρ's (Sec. 5).

4.5 Row V

Row V. Added: the $\log\frac{\vee}{k^+}$ part of line 2, as in Row III. From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right] \tag{4.7}$$

4.6 Row VI

Row VI. Added: the log $\frac{\vee}{k^+}$ part of line 2, as in Row III. From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big] \tag{4.8}$$

5 Four ρ's

Your `BFKL.tex`, unchanged: the three-ρ remainders d^3N_a, d^3N_b (they enter Sec. 6) and the two-ρ lists d^3N_1, d^3N_2. Numerically, $d^3N_1 + d^3N_2$ equals the four-ρ two-ρ terms of the checked sum (`Final_evolution_check.pdf`, Eqs. P12, P13) to 3.2×10^{-15} (charges symmetrized, with c.c.).

$$\frac{d^3N_{a\,\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')\,\alpha$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)\,\alpha'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)\,\alpha'\Big] \tag{5.1}$$

$$\frac{d^3N_{b\,\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)\,\alpha\Big] \tag{5.2}$$

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$$
$$-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$$
$$-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$$
$$-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')$$
$$+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$$
$$-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^{a}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Big]
\end{aligned}
\tag{5.3}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^{e}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^{l}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^{h}(x)\,\rho^{l}(x')\Bigg]
\end{aligned}
\tag{5.4}
$$

6 The three-ρ terms

Same test as in `LL_terms.pdf`, Sec. 7. Every three-ρ term of the list (your lines, the blue lines, your d^3N_a, d^3N_b) is

- renamed to the frame w, w', z, charges x, y, x';
- summed over the 6 placements of the charges and, for Groups I and III, its c.c.;
- evaluated with random points, random SU(3) adjoint Wilson lines ($U^{bc} = U^{cb}$), random charges, and $\log \frac{\vee}{k^+} = 1$.

Your Group I Row II line 2 has the z integral done; it is written back as a z integral. Values at one configuration, in units of $i\mathcal{N}\, e^{-ik\cdot(w'-w)}$:

term	value
BFKL.tex G1 II(2a) line 1	-15.5494
BFKL.tex G1 II(2a) line 2	$+2.9492$
BFKL.tex G1 III N1 line 1	-1.6987
BFKL.tex G1 III N2 line 1	$+3.0986$
BFKL.tex G1 IV line 1	-1.8377
BFKL.tex G3 I line 1	$+1.8377$
BFKL.tex G3 II line 1	$+4.5784$
extracted G1 III N1 line 1	$+45.4745$
extracted G1 III N1 line 2	-45.4745
extracted G1 III N1 line 4	$+3.3973$
extracted G1 III N2 line 1	-19.6585
extracted G1 III N2 line 2	-32.9806
extracted G1 III N2 line 3	$+2.8850$
extracted G1 III N2 line 4	$+2.2837$
extracted G1 III N2 line 5	$+32.9806$
extracted G1 III N2 line 6	$+9.5968$
extracted G1 III N2 line 7	-21.5009
extracted G1 III N2 line 8	$+19.6585$
extracted G1 IV line 1	$+5.3403$
extracted G1 IV line 2	-9.9294
extracted G2 I(1) A line 1	$+1.9040$
extracted G2 I(1) A line 2	$+1.5852$
extracted G2 I(1) A line 3	-1.8952
extracted G2 I(1) B line 1	-2.0102
extracted G2 I(1) B line 2	$+0.0433$
extracted G2 I(1) B line 3	$+5.2217$
extracted G3 I line 1	-5.3403
extracted G3 I line 2	$+9.9294$
extracted G3 II line 1	-4.4229
extracted G3 II line 2	$+5.0271$
extracted G3 III(1) line 2	$+0.3679$
extracted G3 V line 2	$+41.2693$
extracted G3 VI line 2	$+4.1874$
BFKL.tex d3N_a	$+6.3001$
BFKL.tex d3N_b	-1.7932
total	$+4.6e+01$

Largest relative residual over three configurations: 1.01.

- Your `BFKL.tex` alone (without the blue terms): 0.39.
- The three blue line-2 terms of Group III Rows III(1), V, VI alone: 1.11.

7 The two-ρ terms

Every row of the list above equals the $\log \frac{\vee}{k^+}$ part of your rows (check B). The previous list was the same except in four places:

- it had (B), Eq. (7.1), and H_4 (your red line 2 of Group II Row IV);
- it did not have the blue line-2 terms of Group III Rows III(1), III(2), V, VI and Group II Row IV.

Its two-ρ terms (genuine two-ρ terms and the commutator terms of the three-ρ terms, $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$, with all c.c.'s) were added up kernel by kernel and are BFKL⊗LO (`Final_evolution_check.pdf`, `BFKL_final.pdf`). The withdrawn terms (B) were:

$$\text{Row III(1):}\quad -\frac{1}{(2\pi)^3}\,\frac{ig^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$

$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\Big[U^{ac}(w)\,\{\rho^{c}(y),\rho^{b}(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^{d}(y),\rho^{e}(x)\}\Big]$$

Row III(2): $$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}$$
$$\times\left[\frac{(w'-z)\cdot(z-w)}{(w'-z)^2\,(z-w)^2} - \frac{(w'-z)\cdot(x-w)}{2\,(w'-z)^2\,(x-w)^2}\right]$$
$$\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\Big]$$

Row V: $$-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big]$$

Row VI: $$-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big] \tag{7.1}$$

Hence

$$\frac{d^3N}{d^3k}\Big|^{\text{two }\rho}_{\log\frac{\vee}{k^+}} = \mathcal{N}\log\frac{\vee}{k^+}\Big[\text{BFKL}\otimes\text{LO} + \Delta_2\Big],\qquad \Delta_2 = \Big[\text{line 2 of G3 III(1), III(2), V, VI, G2 IV} - (\text{B}) - H_4\Big]_{\text{two }\rho}. \tag{7.2}$$

Δ_2 is computed with the reordering engine of `../two_rho` (the transcriptions of `../bfkl_final/rows.py`). Each transcription equals its LaTeX term here pointwise to 5.0×10^{-14} (`delta_two_rho.py`). Each kernel monomial is evaluated at random points, the colour numerically; colour factors without $U(z)$ are integrated over z in dimensional regularization, $\int_z K(a,b,z) = \pi[P - \log(a-b)^2]$ (MS-bar, with P containing $\frac{2}{\epsilon}$). Δ_2 is non-zero in every one of its 41 structures, for three configurations. Values at one configuration (units of $\mathcal{N}\log\frac{\vee}{k^+}$, x and x' the two charges):

structure	Δ_2	from (blue line 2 of III(1), …, II.IV; B $=-$(B); $H_4=-H_4$)
$-\pi\log(x-w')^2\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	-27.540	III(1), V, VI
$-\pi\log(x-w')^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+10.406$	III(2), V, VI
$-\pi\log(x-w')^2\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	$+11.898$	III(1), V, VI
$-\pi\log(x-w)^2\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	$+29.799$	III(2), V, VI
$-\pi\log(x-w)^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	-16.168	III(1), V, VI
$-\pi\log(x-w)^2\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	$+11.898$	III(1), V, VI
$-\pi\log(x'-w')^2\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	$+34.101$	III(2), V, VI
$-\pi\log(x'-w')^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	$+21.039$	III(1), V, VI
$-\pi\log(x'-w')^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	-16.437	III(1), V, VI
$-\pi\log(x'-w)^2\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	-25.078	III(1), V, VI
$-\pi\log(x'-w)^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	$+21.039$	III(1), V, VI
$-\pi\log(x'-w)^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+11.664$	III(2), V, VI
$-\pi\log(w-w')^2\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	-16.764	III(2), II.IV
$-\pi\log(w-w')^2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	-53.248	III(2), II.IV
$\pi P\;\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)$	$+5.482$	III(1), III(2), V, VI, II.IV
$\pi P\;\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')$	-42.077	III(1), V, VI
$\pi P\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+63.783$	III(1), III(2), V, VI, II.IV
$\pi P\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	-23.795	III(1), V, VI
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$	$+3.924$	III(1), VI
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$	-2.304	III(2), VI
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)$	-1.501	III(2), VI
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)$	-3.105	III(1), VI
$\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$	$+2.986$	III(2)
$\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	$+4.964$	BIII(1), BIII(2), BV, BVI
$\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\frac{1}{(x'-z)^2}$	$+2.288$	BIII(1), BV, BVI
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$	$+2.521$	BIII(1), BIII(2), BV, BVI
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\frac{1}{(x'-z)^2}$	$+6.750$	BIII(1), BV, BVI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)$	-5.018	III(1), VI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)$	-5.018	III(1), VI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)$	$+0.022$	III(2), VI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$	$+3.060$	III(1), VI

$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)$	-1.117	III(1), VI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)$	-0.375	III(2), VI
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$	-1.591	III(2)
$\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	-0.234	III(1), VI
$\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)$	$+0.907$	BIII(1), BIII(2), BV, BVI
$\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\mathcal{K}(x'-w)\cdot\mathcal{K}(w'-z)$	-1.195	BIII(1), BIII(2), BV, BVI
$\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)$	-16.195	BIII(2), H_4
$\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')$	-0.234	III(1), VI
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\frac{1}{(x-z)^2}$	-7.697	BIII(1), BV, BVI
$\mathcal{K}(x'-w)\cdot\mathcal{K}(w'-z)\frac{1}{(x-z)^2}$	-13.648	BIII(1), BV, BVI

So BFKL⊗LO is reproduced only with (B) and H_4 in place of the line-2 terms, i.e. only with the large-p^+ step.

8 Poles

With $E(r)=\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2}{4}$ and $\mathcal{S}$ the LO structure (renaming as in `Final_evolution_check.pdf`, Sec. 6.1):

- **Your two red lines** with a pole are Group I Row V 2b UV and Group II Row IV line 1:

$$-\frac{g^4N_c}{16\pi^4}\big[E(x-w)+E(x'-w')\big]\mathcal{S}\ \text{(with c.c.)}\qquad\text{and}\qquad+\frac{g^4N_c}{8\pi^4}E(w-w')\mathcal{S},$$

  times $\frac{1}{(2\pi)^3k^+}\log\frac{\vee}{k^+}$. Pointwise, their poles cancel (relative 1.5×10^{-15}), but their sum does not (0.23): it is $\frac{g^4N_c}{16\pi^4}\log\frac{(w-w')^4}{(x-w)^2(x'-w')^2}\mathcal{S}$. Neither line is in your rows (check A).

- **The list** has one term with a pole: the blue term of Group II Row IV, $-\frac{g^4N_c}{8\pi^4}E(w-w')\mathcal{S}\times\frac{1}{(2\pi)^3k^+}\log\frac{\vee}{k^+}$. It is minus your red line 1, so its pole is that of the red Group I Row V line (to 1.8×10^{-15}, three configurations), and nothing in the list cancels it.

- In the rows, the other pole terms that depend on $\vee$ are all in the remaining terms:
  - Group I Row II part 1: the pole bracket $\times\log(1-\frac{k^+}{\vee})$;
  - Group II Row IV: the pole bracket $\times$ the ξ integral of line 1 (from $\frac{k^+}{\vee}$), and $\times\log(1-\frac{k^+}{\vee})$ of line 2.

9 Checks

Scripts in `bfkl_rows/`:

check	result
A. `BFKL.tex` (without red) + `Other_terms.tex` = row − JIMWLK, row by row (`check_rows.py`)	$\le 2.5\times10^{-9}$ ($=O(\Lambda)$, $\Lambda=10^{-9}$)
the same with the red lines	G1 V 2b UV, G2 IV: ≥ 0.66
terms with an unevaluated integral: identical text	all
B. BFKL list (black + blue, without red) = $\log\frac{\vee}{k^+}$ part of the rows (exact split, `../llsplit/gen.py`)	$\le 4.4\times10^{-17}$
C. remaining terms = the rest of the rows, $\Lambda\to0$	$\le 3.4\times10^{-16}$
three-ρ terms: list / your `BFKL.tex` alone / line-2 terms alone / second bookkeeping (`three_bfkl.py`)	1.01 / 0.39 / 1.11 / 3.6×10^{-16}
transcriptions of the line-2 terms, (B), H_4 vs the LaTeX (`delta_two_rho.py`)	$\le 5.0\times10^{-14}$
two-ρ change Δ_2: non-zero structures (`delta_two_rho.py`)	41 of 41
four-ρ two-ρ lists vs the checked ones (`compare_4rho.py`)	3.2×10^{-15}
your "BFKL contribution" vs the checked sum of the previous list (`compare_final.py`)	2.1×10^{-16}
red lines: poles / sum; pole of the list vs red G1 V (`pole_bfkl.py`)	1.5×10^{-15} / 0.23; 1.8×10^{-15}